Compressed Quantum Dynamics on Operator Systems and Graph-Cut Correlation Bounds under Tensor-Network Geometry

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Abstract

We study compressed Heisenberg-picture dynamics defined by a unital completely positive compression map $\mathbb{E}_{\mathcal{M}}$ whose image is an operator system $\mathcal{A}_{\mathcal{M}}$ labeled by a tensor-network geometry (G, D). For physical evolutions Λ_k, the manifold-induced channels $\Phi_{\mathcal{M}}^{(k)} = \mathbb{E}_{\mathcal{M}} \circ \Lambda_k$ form a class that is exactly closed under composition, and we bound the difference between step-wise compression and compression applied once at the end of the evolution. The bound is stated pointwise on propagated observables and is informative only when these remain close to $\mathcal{A}_{\mathcal{M}}$.

Tensor-network geometry is connected to this structure through an explicit representability hypothesis: every state on $\mathcal{A}_{\mathcal{M}}$ lies within η_D of the restriction of a tensor-network state with graph G and bond dimension at most D. Under this hypothesis the standard Schmidt-rank argument gives a graph-cut mutual-information bound for the representatives and a geometric capacity functional $2 \log D \, |\partial_G|$. These geometric statements are conditional on the hypothesis: the geometry enters through the hypothesis and the admissibility condition on $\mathbb{E}_{\mathcal{M}}$, and is not derived from the operator-algebraic layer. We do not construct a non-trivial admissible compression map; constructing one and deriving representability are left as open problems.

1 Introduction

Tensor-network methods have become central to the study of interacting quantum many-body systems, enabling efficient representations of highly correlated states through compressed variational manifolds such as matrix product states (MPS), projected entangled pair states (PEPS), and multiscale entanglement renormalization ansätze (MERA) [33, 29, 31, 23, 32]. These approaches exploit physically motivated entanglement structure to approximate relevant sectors of exponentially large Hilbert spaces while maintaining computational tractability. Beyond their numerical utility, tensor-network manifolds encode geometric and correlation constraints that strongly influence accessible many-body dynamics.

In parallel, quantum information theory studies the operational behavior of quantum channels, adaptive protocols, and entanglement propagation under noisy and memory-dependent evolution [22]. Existing connections between tensor networks and quantum channels include transfer-operator constructions, matrix-product operator (MPO) representations of open-system evolution, tensor-network descriptions of quantum circuits, and process-tensor approaches to non-Markovian dynamics. However, these frameworks typically interpret tensor networks as representations of states, operators, or histories rather than as compression structures that directly constrain achievable information-processing tasks.

In this work, we introduce a Tensor-Network Channel (TNC) formalism that associates variational tensor-network geometries with effective observable subsystems and manifold-induced quantum channels. The central object is a tensor-network compression map

$$\mathbb{E}_{\mathcal{M}} : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H}),$$

whose image defines an operator system representing the observables retained by a tensor-network description of bounded bond dimension. Given a physical Heisenberg-picture evolution

$$\Lambda : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H}),$$

the associated manifold-induced channel is defined by

$$\Phi_{\mathcal{M}} = \mathbb{E}_{\mathcal{M}} \circ \Lambda.$$

The resulting channel takes values in the operator system $\mathcal{A}_{\mathcal{M}}$; compression-induced information loss is quantified by the compression defect introduced in Sec. 3.

The compression map is also required to satisfy an approximate idempotence condition, reflecting the intuition that repeated compression should not substantially alter an already compressed description. This condition is part of the definition of admissibility; none of the results below use it.

The central perspective developed here is that tensor-network geometry admits a natural operational interpretation at the level of observables. In the present framework, finite bond dimension and network structure are assumed (through the admissibility condition on E) to determine which observables are retained after compression, and thereby constrain the effective correlations and information flow that can be represented within the compressed theory. Rather than treating a tensor-network manifold as a dynamical state space, we regard it as inducing an effective observable system together with a corresponding class of manifold-constrained quantum channels.

We show that the channel class generated by $\mathbb{E}_{\mathcal{M}}$ is exactly closed under composition, and we bound the difference between step-wise compression and compression applied once at the end of a sequence of evolutions. Tensor-network geometry is then connected to the compressed dynamics through an explicit representability hypothesis, under which standard graph-cut entropy bounds apply to the representatives.

Conceptually, the TNC framework differs from conventional transfer-operator, matrix-product-operator, and process-tensor approaches by treating tensor-network geometry as the source of an effective compression structure on the observable algebra. The induced channels arise from observable compression rather than from projected state trajectories. Consequently, finite-bond tensor-network structure acquires an operational meaning: it is assumed to determine which observables, correlations, and information-processing capabilities remain accessible within the compressed theory.

A distinguishing feature of the present framework is that all fundamental dynamical objects are formulated in the Heisenberg picture. State-space descriptions enter only through the Schrödinger duals of the induced channels when entropy, correlation, or entanglement measures are evaluated. This separation permits tensor-network geometry to be incorporated into a fully composable channel-theoretic framework while retaining compatibility with standard information-theoretic quantities.

The remainder of this work is organized as follows. In Sec. 2 we review variational tensor-network manifolds. In Sec. 3 we define compression maps, operator systems and manifold-induced channels, show that the channel class is closed under composition, and prove a stability bound for step-wise compression. In Sec. 4 we state the representability hypothesis. In Secs. 5 and 6 we derive the graph-cut mutual-information bound and the capacity functional for representatives. Sec. 7 compares the capacity functional across architectures, Sec. 8 discusses scope and limitations, Sec. 9 states open problems, and Sec. 10 concludes. Appendices A and B contain the proofs of Proposition 1 and Lemma 1.

2 Variational Manifolds and Tensor-Network Structure

Let $\mathcal{H}$ denote the Hilbert space of an interacting quantum lattice system. A variational tensor-network manifold

$$\mathcal{M} \subset \mathcal{H}$$

is specified by a family of parameterized states

$$|\psi(\theta)\rangle \in \mathcal{M},$$

where

$$\theta \in \Theta$$

belongs to a finite-dimensional parameter space.

In the present work, the manifold itself is not regarded as a dynamical state space. Rather, it serves as the geometric structure from which a tensor-network compression framework will be constructed.

Examples include:

- Matrix Product States (MPS),
- Projected Entangled Pair States (PEPS),
- MERA networks.

Each such tensor-network family is naturally associated with an underlying graph structure G together with a bond-dimension parameter D. These geometric data will serve as the primary descriptors of tensor-network complexity throughout this work.

Tensor-network representations encode correlations through graph connectivity and bond dimensions D. The pair

$$(G, D)$$

determines the effective complexity of the tensor-network description and constrains the correlations and entanglement patterns that can be represented within the variational family. For a bipartition $A|B$, the entanglement entropy of a tensor-network state is bounded by

$$S(A) \leq N_{\mathrm{cut}} \log D, \tag{1}$$

where N_{cut} denotes the number of virtual bonds intersected by the bipartition and D is the bond-dimension bound, up to geometry-dependent constants associated with the number of bonds crossing the bipartition.

For a left/right bipartition of an open-boundary matrix product state one has $N_{\mathrm{cut}} = 1$ (and $N_{\mathrm{cut}} \leq 2$ for a contiguous block), recovering the familiar bound

$$S(A) \leq \log D.$$

For more general tensor-network geometries such as PEPS and MERA, the value of N_{cut} depends on the network structure and the chosen bipartition.

These observations suggest that tensor-network geometry may be viewed not only as a variational approximation scheme but also as a source of operational constraints on observable information. Rather than describing dynamics directly on the manifold $\mathcal{M}$, the framework developed in the next section associates to the tensor-network geometry an effective compression structure on the observable algebra $\mathcal{B}(\mathcal{H})$. The resulting compressed observable theory provides the basis for the manifold-induced quantum channels introduced later.

3 Manifold-Induced Quantum Channels

We now define the central operational object of this work. Throughout, we work in the Heisenberg picture, so physical evolutions are represented by unital completely positive maps on operator algebras. We begin with a tensor-network compression structure defined by an equivalence class of admissible unital completely positive maps

$$E \in \mathfrak{C}(G, D),$$

where $\mathfrak{C}(G, D)$ denotes the set of all admissible compression maps associated with tensor-network geometry specified by graph structure G and bond dimension bound D. Once a compression map

$$E \in \mathfrak{C}(G, D)$$

has been fixed, we denote it by

$$\mathbb{E}_{\mathcal{M}}$$

for notational convenience. Below, E and $\mathbb{E}_{\mathcal{M}}$ denote the same fixed map. Each element of $\mathfrak{C}(G, D)$ is a unital completely positive map

$$E : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$$

whose image defines an operator system associated with tensor-network representable observables compatible with bond dimension D. Its precise structural requirements are specified in the following definition.

An admissible compression map $E \in \mathfrak{C}(G, D)$ is required to satisfy:

1. $\mathrm{Im}(E)$ is an operator system in $\mathcal{B}(\mathcal{H})$;
2. $\|E^2 - E\|_\diamond \leq \varepsilon_D$;
3. the image $\mathrm{Im}(E)$ admits tensor-network realizations compatible with graph structure G and bond dimension bounded by D;
4. for a family $\{E_D\}_D$ one may ask that $\varepsilon_D \to 0$ as $D \to \infty$; this is a property of the family, no such family is constructed here, and no result below uses it.

The present framework is intentionally formulated at the level of equivalence classes rather than canonical constructions. We do not assume a unique tensor-network induced projection map. Instead, $\mathfrak{C}(G, D)$ represents the class of all admissible coarse-graining maps compatible with the tensor-network geometry (G, D). All subsequent results are required to hold uniformly for any choice

$$E \in \mathfrak{C}(G, D)$$

.

Different tensor-network constructions may give rise to different admissible compression maps. The results of this work depend only on the structural properties stated below.

The primary object of the framework is the tensor-network compression structure

$$(E, \mathcal{A}_{\mathcal{M}}), \quad E \in \mathfrak{C}(G, D),$$

together with the induced operator system

$$\mathcal{A}_{\mathcal{M}} := \mathrm{Im}(E).$$

rather than the tensor-network geometry alone. The geometry (G, D) serves to motivate and constrain admissible compression maps, but uniqueness of the resulting compression structure is neither assumed nor required.

Unless stated otherwise, all constructions are formulated relative to a fixed choice

$$E \in \mathfrak{C}(G, D),$$

with associated operator system

$$\mathcal{A}_{\mathcal{M}} = \mathrm{Im}(E).$$

Whenever a statement is asserted as a property of the Tensor-Network Channel framework itself, it is intended to depend only on the structural properties shared by all admissible choices

$$E \in \mathfrak{C}(G, D).$$

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3.1 Tensor-Network Operator Systems

Throughout this work, all dynamical objects are formulated in the Heisenberg picture. Let $\mathcal{H}$ be a finite-dimensional Hilbert space and let

$$\mathcal{B}(\mathcal{H})$$

denote the full operator algebra.

Definition 1 (Tensor-Network Operator System)**.** *Let*

$$\mathbb{E}_{\mathcal{M}} : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$$

be a unital completely positive map associated with a tensor-network geometry specified by graph structure G and bond dimension bound D. The associated tensor-network operator system is defined by

$$\mathcal{A}_{\mathcal{M}} := \mathrm{Im}(\mathbb{E}_{\mathcal{M}}).$$

We restrict attention to maps whose images form operator systems. Accordingly, $\mathcal{A}_{\mathcal{M}}$ is assumed to be a self-adjoint linear subspace of $\mathcal{B}(\mathcal{H})$ containing the identity operator. Operationally, $\mathcal{A}_{\mathcal{M}}$ represents the collection of effective observables whose expectation values are retained under tensor-network compression at bond dimension D. No multiplicative closure is assumed. In particular,

$$XY \notin \mathcal{A}_{\mathcal{M}}$$

need not hold even when

$$X, Y \in \mathcal{A}_{\mathcal{M}}.$$

The tensor-network operator system is therefore regarded as an operator-system object rather than a C^*-subalgebra of $\mathcal{B}(\mathcal{H})$.

The operator system $\mathcal{A}_{\mathcal{M}}$ serves as the fundamental observable space of the Tensor-Network Channel framework. All Heisenberg evolutions, compression maps, and operational quantities are defined relative to this operator system.

We emphasize that the class $\mathfrak{C}(G, D)$ is assumed to be nonempty for all admissible tensor-network geometries (G, D). No uniqueness or canonical selection of $E \in \mathfrak{C}(G, D)$ is assumed. The role of the tensor-network geometry is to constrain admissible coarse-graining maps rather than to determine a unique projection.

The existence of admissible compression maps is treated as a structural assumption of the framework. Establishing explicit constructions for particular tensor-network architectures lies beyond the scope of the present work.

Definition 2 (Tensor-Network Compression Map)**.** *Let:*

- $\mathcal{H}$ *denote the physical Hilbert space,*
- $\mathcal{B}(\mathcal{H})$ *denotes the bounded operators on* $\mathcal{H}$*,*
- $\mathcal{A}_{\mathcal{M}} \subseteq \mathcal{B}(\mathcal{H})$ *denote the tensor-network operator system associated with the manifold* $\mathcal{M}$*, with graph structure* G *and bond dimension* D*,*
- $\Lambda : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$ *denote a unital completely positive physical evolution.*

We define a tensor-network compression map

$$\mathbb{E}_{\mathcal{M}} : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H}),$$

to be a unital completely positive map whose image

$$\mathrm{Im}(\mathbb{E}_{\mathcal{M}}) = \mathcal{A}_{\mathcal{M}}$$

defines the tensor-network operator system associated with the geometry (G, D)*, satisfying approximate idempotence*

$$\|\mathbb{E}_{\mathcal{M}}^2 - \mathbb{E}_{\mathcal{M}}\|_\diamond \leq \varepsilon_D,$$

where $\varepsilon_D \geq 0$ *is the idempotence defect (for a family* $\{E_D\}$ *one may ask* $\varepsilon_D \to 0$*; this is not used below). In addition, the image*

$$\mathcal{A}_{\mathcal{M}} = \mathrm{Im}(\mathbb{E}_{\mathcal{M}})$$

is assumed to admit tensor-network realizations compatible with graph structure G *and bond dimension bounded by* D*, in the sense specified by the admissibility conditions defining* $\mathfrak{C}(G, D)$*. Thus the compression map identifies the effective observable subsystem that is representable within the tensor-network geometry. The associated manifold-induced channel (Heisenberg evolution) is then defined by*

$$\Phi_{\mathcal{M}} = \mathbb{E}_{\mathcal{M}} \circ \Lambda. \tag{2}$$

This map acts in the Heisenberg picture on observables. All state-dependent quantities (including entropies, mutual informations, and entanglement measures) are defined only via the Schrödinger dual map $\Phi_{\mathcal{M}}^*$*, which is induced by* $\Phi_{\mathcal{M}}$ *through*

$$\mathrm{Tr}[\Phi_{\mathcal{M}}(X)\rho] = \mathrm{Tr}[X\, \Phi_{\mathcal{M}}^*(\rho)].$$

The tensor-network compression map

$$\mathbb{E}_{\mathcal{M}}$$

is the primary structural object of the framework. Its image

$$\mathcal{A}_{\mathcal{M}} = \mathrm{Im}(\mathbb{E}_{\mathcal{M}})$$

defines the effective observable subsystem retained by tensor-network compression. The pair

$$(\mathbb{E}_{\mathcal{M}}, \mathcal{A}_{\mathcal{M}})$$

will be referred to as a tensor-network compression structure. No assumption is made that

$$\mathcal{A}_{\mathcal{M}}$$

is a C*-subalgebra or that

$$\mathbb{E}_{\mathcal{M}}$$

is a conditional expectation in the operator-algebraic sense. Only complete positivity, unitality, approximate idempotence, and tensor-network realizability of the image are required.

Operationally,

$$\mathbb{E}_{\mathcal{M}}$$

acts as an observable compression operator. Given an observable

$$X \in \mathcal{B}(\mathcal{H}),$$

the compressed observable

$$\mathbb{E}_{\mathcal{M}}(X)$$

represents the effective information about X retained within the tensor-network geometry. The quantity

$$X - \mathbb{E}_{\mathcal{M}}(X)$$

therefore measures information discarded by the tensor-network description. In this sense, tensor-network geometry induces an effective observable theory inside

$$\mathcal{B}(\mathcal{H}),$$

and manifold-induced channels describe physical evolutions followed by compression into this effective observable subsystem.

Its action on states is defined only through the dual map $\mathbb{E}^*_{\mathcal{M}}$ when state-dependent quantities are required. Unlike nonlinear variational projection schemes, the compression-map formulation preserves complete positivity and compositional structure, enabling a consistent operational algebra of manifold-constrained quantum channels.

This construction transforms variational geometry into a composable operational object.

Operational interpretation. The formalism developed in this work does not track the exact physical state trajectory as a fundamental object. Instead, it defines a *compressed (or shadow) dynamics* on the operator algebra induced by the tensor-network projection map $\mathbb{E}_{\mathcal{M}}$. All physically relevant state-dependent quantities are understood as arising from the Schrödinger dual of this Heisenberg-picture compression semigroup, and are therefore constrained by the geometry of the induced operator subsystem rather than by the full Hilbert-space evolution.

The framework is therefore fundamentally Heisenbergian. The tensor-network manifold itself does not constitute a dynamical state space. Rather, it determines an effective observable subsystem through the compression map

$$\mathbb{E}_{\mathcal{M}}.$$

All subsequent constructions are formulated at the level of compressed observables, while state-space descriptions enter only through duality when information-theoretic quantities are evaluated.

Definition 3 (Intrinsic State Space)**.** *Let*

$$\mathcal{A}_{\mathcal{M}} = \mathrm{Im}(\mathbb{E}_{\mathcal{M}})$$

be the tensor-network operator system. The intrinsic state space associated with the tensor-network compression structure is

$$\mathfrak{S}(\mathcal{A}_{\mathcal{M}}) := \{\omega : \mathcal{A}_{\mathcal{M}} \to \mathbb{C} \mid \omega \text{ is linear, positive, and } \omega(I) = 1\}.$$

Every physical density operator

$$\rho \in \mathcal{D}(\mathcal{H})$$

induces an intrinsic state

$$\omega_\rho(X) = \mathrm{Tr}(X\rho), \qquad X \in \mathcal{A}_{\mathcal{M}}.$$

The framework therefore identifies states only through their action on the compressed observable subsystem.

3.2 Heisenberg–Schrödinger Duality (Derived Structure)

All dynamical objects in this framework are defined in the Heisenberg picture. For any unital completely positive map

$$\Phi : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H}),$$

its Schrödinger dual

$$\Phi^*$$

is the unique trace-dual map satisfying

$$\mathrm{Tr}[\Phi(X)\rho] = \mathrm{Tr}[X\,\Phi^*(\rho)]$$

for all $X \in \mathcal{B}(\mathcal{H})$ and $\rho \in \mathcal{D}(\mathcal{H})$. Because Φ is unital and completely positive, its dual Φ^* is completely positive and trace preserving.

In particular, the tensor-network compression map $\mathbb{E}_{\mathcal{M}}$ induces a dual action $\mathbb{E}^*_{\mathcal{M}}$ on states, which is used only for the evaluation of entropy and correlation measures. No physical postulate is assigned to state evolution independently of Heisenberg dynamics.

3.3 Tensor-Network Channel Algebra

We now formalize the algebraic structure induced by manifold-constrained dynamics.

Definition 4 (Tensor-Network Channel Class)**.** *Fix an admissible compression structure*

$$(\mathbb{E}_{\mathcal{M}}, \mathcal{A}_{\mathcal{M}}), \qquad \mathbb{E}_{\mathcal{M}} \in \mathfrak{C}(G, D).$$

Let $\mathcal{M} \subset \mathcal{H}$ be a variational tensor-network manifold. The associated Tensor-Network Channel (TNC) class is defined as

$$\mathsf{TNC}(\mathcal{M}) = \{\Phi_{\mathcal{M}} = \mathbb{E}_{\mathcal{M}} \circ \Lambda \mid \Lambda \in \mathrm{UCP}(\mathcal{B}(\mathcal{H}))\},$$

where:

- Λ *is a unital completely positive (UCP) physical evolution,*
- $\mathbb{E}_{\mathcal{M}}$ *denotes the tensor-network compression map surjective onto the operator subsystem $\mathcal{A}_{\mathcal{M}}$.*

Different admissible compression maps may generate different Tensor-Network Channel classes. All structural results below are formulated relative to a fixed compression structure and depend only on the admissibility properties shared by all elements of $\mathfrak{C}(G, D)$.

Since compositions of UCP maps remain UCP, every element of $\mathsf{TNC}(\mathcal{M})$ defines a UCP map on $\mathcal{B}(\mathcal{H})$.

Observation 1 (Exact Closure of the Channel Class)**.** $\mathsf{TNC}(\mathcal{M})$ *is closed under composition. For $\Phi_i = \mathbb{E}_{\mathcal{M}} \circ \Lambda_i$,*

$$\Phi_2 \circ \Phi_1 = \mathbb{E}_{\mathcal{M}} \circ \big(\Lambda_2 \circ \mathbb{E}_{\mathcal{M}} \circ \Lambda_1\big),$$

and $\Lambda_2 \circ \mathbb{E}_{\mathcal{M}} \circ \Lambda_1 \in \mathrm{UCP}(\mathcal{B}(\mathcal{H}))$. Hence $\Phi_2 \circ \Phi_1 \in \mathsf{TNC}(\mathcal{M})$ with no defect, and without using idempotence.

Definition 5 (Compression Defect)**.** *For a physical evolution*

$$\Lambda \in \mathrm{UCP}(\mathcal{B}(\mathcal{H})),$$

the compression defect is

$$d_{\mathcal{M}}(\Lambda) := \|\Lambda - \mathbb{E}_{\mathcal{M}} \circ \Lambda\|_{\diamond}.$$

Definition 6 (Leakage Kernel)**.** *The leakage kernel associated with the compression structure*

$$(\mathbb{E}_{\mathcal{M}}, \mathcal{A}_{\mathcal{M}})$$

is the collection

$$\mathcal{I}_{\mathcal{M}} = \ker(\mathbb{E}_{\mathcal{M}}),$$

consisting of operators annihilated by tensor-network compression. The associated leakage magnitude is quantified by the compression defect $d_{\mathcal{M}}(\Lambda)$*.*

Operators in $\mathcal{I}_{\mathcal{M}} = \ker(\mathbb{E}_{\mathcal{M}})$ are annihilated by the compression map and therefore have no representation within the compressed observable subsystem $\mathcal{A}_{\mathcal{M}}$. Consequently, only their compressed images contribute to expectation values computed within the Heisenberg-picture compressed theory, and these contributions vanish by construction. The leakage kernel therefore represents information discarded by tensor-network compression.

The TNC framework therefore consists of:

- manifold-induced quantum channels,
- approximate composition rules controlled by compression defects,
- leakage structures associated with tensor-network compression,
- information-theoretic quantities derived from the compressed dynamics.

This promotes tensor-network geometry into an operational algebraic structure governing adaptive many-body quantum dynamics.

3.4 Optional Tensor-Network Realization

The Tensor-Network Channel framework developed in the preceding sections is formulated entirely at the level of operator systems and manifold-induced quantum channels.

Its fundamental objects are the compression structure

$$(\mathbb{E}_{\mathcal{M}}, \mathcal{A}_{\mathcal{M}}),$$

the associated intrinsic state space

$$\mathfrak{S}(\mathcal{A}_{\mathcal{M}}),$$

and the induced Heisenberg dynamics generated by manifold-induced channels.

No tensor-network state representative is required for the definition of any of these objects. In particular, the framework does not assume that intrinsic states

$$\omega \in \mathfrak{S}(\mathcal{A}_{\mathcal{M}})$$

arise from tensor-network states of bounded bond dimension, nor does it require the existence of a reconstruction map from intrinsic states to tensor-network state representatives.

Nevertheless, one may ask whether intrinsic states generated by the compressed dynamics admit compatible tensor-network realizations. Such realizations provide a connection between the operator-algebraic description developed here and the geometric structure of tensor networks.

The existence of tensor-network representatives is not assumed in the construction of the Tensor-Network Channel framework itself. Whenever such realizations are required, they will be introduced through an additional representability hypothesis.

This additional structure is developed in the next section, where tensor-network realizations are treated as optional geometric representatives of the compressed operator dynamics rather than as fundamental ingredients of the framework.

3.5 Remark on Leakage

Since $\Phi_{\mathcal{M}} = \mathbb{E}_{\mathcal{M}} \circ \Lambda$ maps all of $\mathcal{B}(\mathcal{H})$ into $\mathcal{A}_{\mathcal{M}}$, every element of $\mathsf{TNC}(\mathcal{M})$ leaves $\mathcal{A}_{\mathcal{M}}$ invariant exactly. The only error in the framework is the compression defect $d_{\mathcal{M}}(\Lambda)$; no separate leakage assumption is needed.

Proposition 1 (Stability of Step-wise Compression)**.** *Let $E := \mathbb{E}_{\mathcal{M}}$, let $\Lambda_1, \dots, \Lambda_n \in \mathrm{UCP}(\mathcal{B}(\mathcal{H}))$ and $\Phi_{\mathcal{M}}^{(k)} = E \circ \Lambda_k$. Then for every $X \in \mathcal{B}(\mathcal{H})$,*

$$\big\| \big(\Phi_{\mathcal{M}}^{(n)} \circ \cdots \circ \Phi_{\mathcal{M}}^{(1)} - E \circ \Lambda_n \circ \cdots \circ \Lambda_1\big)(X) \big\| \le \sum_{j=1}^{n-1} \big\| (\mathrm{id} - E)\big(\Lambda_j \cdots \Lambda_1(X)\big) \big\|.$$

Consequently

$$\Delta_n := \big\| \Phi_{\mathcal{M}}^{(n)} \circ \cdots \circ \Phi_{\mathcal{M}}^{(1)} - E \circ \Lambda_n \circ \cdots \circ \Lambda_1 \big\|_\diamond \le \sum_{j=1}^{n-1} d_{\mathcal{M}}(\Lambda_j), \qquad d_{\mathcal{M}}(\Lambda) := \|\Lambda - E \circ \Lambda\|_\diamond.$$

In particular $\Delta_1 = 0$.

Proof idea. Write $A_j = E\Lambda_n E\Lambda_{n-1} \cdots E\Lambda_j \Lambda_{j-1} \cdots \Lambda_1$. Then A_1 is the step-wise compressed evolution, $A_n = E\Lambda_n \cdots \Lambda_1$, and $A_j - A_{j+1} = P_j\,(E - \mathrm{id})\Lambda_j\, Q_j$ with P_j, Q_j unital completely positive. Telescoping over $j = 1, \dots, n-1$ and using that UCP maps are contractions gives the bound; the full proof is in Appendix A. Since both maps are UCP, $\Delta_n \le 2$ trivially, so the bound is informative only when its right-hand side is small, i.e. when the propagated observables $\Lambda_j \cdots \Lambda_1(X)$ stay close to $\mathcal{A}_{\mathcal{M}}$. Idempotence of E is not used.

Proposition 1 compares compressing after every step with compressing once at the end. It is not needed for closure, which holds exactly (Observation 1), and it is not used in Secs. 5–6.

Convention on ordering. For Heisenberg-picture maps, $(\Psi_1 \circ \cdots \circ \Psi_k)^* = \Psi_k^* \circ \cdots \circ \Psi_1^*$. Throughout Secs. 4–6, the state $\rho_k = \Phi_{\mathcal{M}}^{(k)*} \circ \cdots \circ \Phi_{\mathcal{M}}^{(1)*}(\rho_0)$ is therefore the Schrödinger dual of the Heisenberg composition $\Phi_{\mathcal{M}}^{(1)} \circ \cdots \circ \Phi_{\mathcal{M}}^{(k)}$, in which step 1 acts first on the state. Proposition 1 applies to this composition after reversing the labels of the Λ_j.

4 Representability Structure and Effective Tensor-Network Realizations

The operator-algebraic framework developed in Sections 2 and 3 is formulated entirely in terms of tensor-network compression structures

$$(\mathbb{E}_{\mathcal{M}}, \mathcal{A}_{\mathcal{M}}),$$

without requiring explicit tensor-network state representatives.

Throughout this section, the graph structure G and bond-dimension bound D are those associated with the fixed compression structure $(\mathbb{E}_{\mathcal{M}}, \mathcal{A}_{\mathcal{M}})$.

To relate compressed Heisenberg dynamics to tensor-network geometry, we introduce an additional representability assumption. This assumption is not required for the construction of the Tensor-Network Channel (TNC) algebra itself, but is used in subsequent sections to connect compressed operator dynamics with geometric correlation bounds.

Here a tensor-network state with graph structure G and bond dimension bounded by D means any density operator admitting a tensor-network representation compatible with G whose virtual bond dimensions do not exceed D.

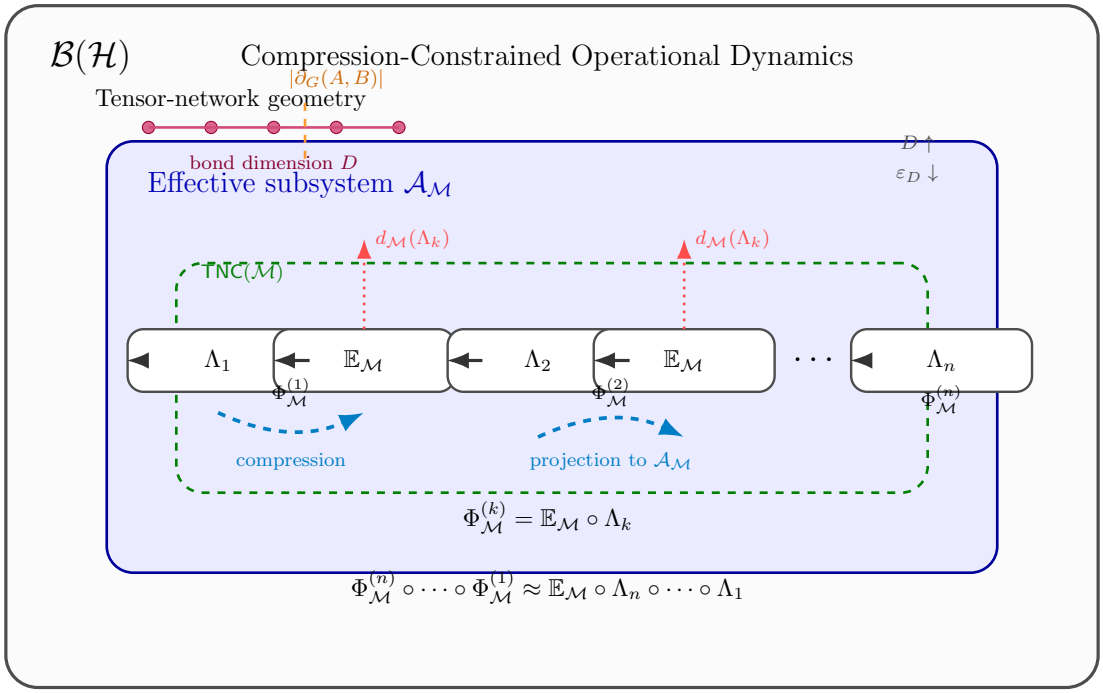


Figure 1: Operational structure of manifold-induced quantum channels. Physical evolutions Λ_k act on the full operator algebra $\mathcal{B}(\mathcal{H})$, while tensor-network compression maps $\mathbb{E}_\mathcal{M}$ project dynamics onto the effective operator subsystem $\mathcal{A}_\mathcal{M}$ induced by the tensor-network geometry. Sequential compositions define manifold-induced channels in $\mathsf{TNC}(\mathcal{M})$, which is closed under composition; the difference from compressing once at the end is controlled by $d_\mathcal{M}(\Lambda_k)$ (Proposition 1). Under the representability hypothesis, graph cuts and bond dimension bound the correlations of the representatives.

Assumption 1 (Representability Structure)**.** *Let*

$$(\mathbb{E}_\mathcal{M}, \mathcal{A}_\mathcal{M})$$

be a tensor-network compression structure. Let

$$\mathfrak{S}_{\mathrm{rest}}(\mathcal{A}_\mathcal{M}) := \{\omega_\rho \mid \rho \in \mathcal{D}(\mathcal{H}),\ \omega_\rho(X) = \mathrm{Tr}(X\rho) \quad \forall X \in \mathcal{A}_\mathcal{M}\}$$

be the set of restrictions of density operators to $\mathcal{A}_\mathcal{M}$. By the Arveson–Krein extension theorem [24] this is the full state space $\mathfrak{S}(\mathcal{A}_\mathcal{M})$; in particular it contains every state produced by any sequence of manifold-induced channels, so the assumption below imposes no dynamical restriction. Assume that for every

$$\omega \in \mathfrak{S}_{\mathrm{rest}}(\mathcal{A}_\mathcal{M}),$$

there exists a tensor-network state

$$\widetilde{\rho}$$

with graph structure G and bond dimension bounded by D such that

$$\sup_{\substack{X \in \mathcal{A}_\mathcal{M} \\ \|X\| \le 1}} |\omega(X) - \mathrm{Tr}(X\widetilde{\rho})| \le \eta_D.$$

No uniqueness is assumed. We further assume that

$$\eta_D \to 0 \qquad as \qquad D \to \infty.$$

The representability structure is an additional hypothesis relating the compressed observable subsystem to tensor-network geometry. It is not derived from the operator-algebraic framework developed previously and should be regarded as an independent structural assumption. It is a static hypothesis on $\mathcal{A}_{\mathcal{M}}$ alone: it makes no reference to the dynamics, and the tensor-network geometry in the conclusions of Secs. 5 and 6 is the geometry assumed here.

Definition 7 (Effective Tensor-Network Realization Set)**.** *Let*

$$(\mathbb{E}_{\mathcal{M}}, \mathcal{A}_{\mathcal{M}})$$

be a tensor-network compression structure satisfying the representability assumption. The effective tensor-network realization set

$$\mathcal{M}_{\mathrm{eff}}(D)$$

is the collection of all tensor-network states $\widetilde{\rho}$ with bond dimension bounded by D for which there exists an intrinsic state

$$\omega \in \mathfrak{S}_{\mathrm{rest}}(\mathcal{A}_{\mathcal{M}})$$

satisfying

$$\sup_{\substack{X \in \mathcal{A}_{\mathcal{M}} \\ \|X\| \leq 1}} |\omega(X) - \mathrm{Tr}(X\widetilde{\rho})| \leq \eta_D.$$

Under the representability assumption, every dual-evolved state

$$\rho_k = \left(\Phi_{\mathcal{M}}^{(k)*} \circ \cdots \circ \Phi_{\mathcal{M}}^{(1)*}\right)(\rho_0)$$

determines an intrinsic state

$$\omega_k := \omega_{\rho_k} \in \mathfrak{S}_{\mathrm{rest}}(\mathcal{A}_{\mathcal{M}})$$

through restriction to the compressed operator subsystem,

$$\omega_k(X) = \mathrm{Tr}(X\rho_k), \qquad X \in \mathcal{A}_{\mathcal{M}}.$$

By assumption, there therefore exists a representative tensor-network state

$$\widetilde{\rho}_k \in \mathcal{M}_{\mathrm{eff}}(D)$$

such that

$$\sup_{\substack{X \in \mathcal{A}_{\mathcal{M}} \\ \|X\| \leq 1}} |\mathrm{Tr}(X\rho_k) - \mathrm{Tr}(X\widetilde{\rho}_k)| \leq \eta_D.$$

Thus every compressed dynamical state admits a tensor-network realization at the level of the accessible observable subsystem.

Role in the framework. Sections 2 and 3 establish the Tensor-Network Channel algebra independently of any tensor-network state realization.

The representability assumption introduced here provides an optional geometric layer that allows operator-algebraic dynamics to be interpreted through tensor-network representatives.

All subsequent correlation and entropy bounds are derived under this assumption and should therefore be understood as consequences of the operator-algebraic framework together with the existence of compatible tensor-network realizations.

5 Tensor-Network Representation of Operational Dynamics

Tensor-network geometry naturally induces constraints on information propagation. We associate:

- network edges with transmissive correlation pathways,
- bond dimensions with finite correlation capacity,
- network depth with adaptive propagation complexity.

This motivates a graph-theoretic operational interpretation of tensor networks.

Lemma 1 (Bond-Dimension Information Bound)**.** *Let $|\psi\rangle \in \mathcal{M}_{\mathrm{eff}}(D)$ be a pure tensor-network state with graph structure G and maximal bond dimension D. Assume that the tensor-network architecture satisfies the standard graph-cut Schmidt-rank property: every edge crossing a bipartition contributes Schmidt rank at most D. Then for any bipartition $A|B$ of the whole system (so that $A \cup B$ is the entire system and ρ_ψ is pure on AB),*

$$I(A:B)_\psi \leq 2 \log D \cdot |\partial_G(A,B)|,$$

where $\rho_\psi = |\psi\rangle\langle\psi|$ and ρ_A, ρ_B denote the reduced density matrices of ρ_ψ on subsystems A and B, respectively,

$$I(A:B)_\psi = S(\rho_A) + S(\rho_B) - S(\rho_{AB}).$$

denotes the quantum mutual information of $|\psi\rangle$, and $|\partial_G(A,B)|$ denotes the number of edges in a minimal edge cut of G separating A and B.

The proof is given in Appendix B.

Interpretation. Lemma 1 is a standard graph-cut consequence of finite bond dimension. It expresses an operational form of a tensor-network area law [10]: the geometric upper bound on bipartite correlations is controlled by the minimal graph cut rather than by the full Hilbert-space dimension.

For acyclic tensor-network geometries, including matrix-product states and tree tensor networks, the bound follows directly from Schmidt-rank constraints across graph cuts. For architectures containing loops, such as projected entangled-pair states, the same expression should be interpreted as a geometric upper bound associated with finite bond dimension rather than as a universal equality.

5.1 Operational Consequence under the Representability Assumption

The representability assumption introduced in Section 4 provides a bridge between the compressed operator-system dynamics and tensor-network geometry. Consider a sequential dual evolution

$$\rho_k = \left(\Phi_{\mathcal{M}}^{(k)*} \circ \cdots \circ \Phi_{\mathcal{M}}^{(1)*}\right)(\rho_0).$$

Let

$$\omega_k = \rho_k|_{\mathcal{A}_\mathcal{M}} \in \mathfrak{S}_{\mathrm{rest}}(\mathcal{A}_\mathcal{M})$$

denote the corresponding intrinsic state on the compressed operator subsystem.

Under the representability assumption of Section 4, there exists a tensor-network representative

$$\widetilde{\rho}_k \in \mathcal{M}_{\mathrm{eff}}(D)$$

such that

$$\sup_{\substack{X\in\mathcal{A}_{\mathcal{M}}\\ \|X\|\le 1}} |\omega_k(X) - \mathrm{Tr}(X\widetilde{\rho}_k)| \le \eta_D.$$

Consequently, information-theoretic quantities evaluated on a chosen tensor-network representative inherit the geometric constraints imposed by the underlying tensor-network architecture, although these constraints are properties of the representative realization and not of the compressed operator dynamics alone.

In particular, if a representative $\widetilde{\rho}_k \in \mathcal{M}_{\mathrm{eff}}(D)$ is chosen to admit a pure tensor-network purification satisfying the graph-cut Schmidt-rank property of Lemma 1, then one obtains

$$I(A:B)_{\widetilde{\rho}_k} \le 2\log D \cdot |\partial_G(A,B)|.$$

Thus every tensor-network realization provided by the representability assumption obeys a geometric correlation constraint determined by the graph structure and bond dimension. In this sense, tensor-network geometry imposes an effective limitation on the correlation structure represented within the compressed observable subsystem.

The graph-cut quantity

$$|\partial_G(A,B)|$$

therefore serves as a natural geometric measure of the correlation capacity available within the tensor-network compression structure.

This observation motivates the capacity functional introduced in the next section.

6 Consequences of Representability for Correlations and Capacity

This section derives geometric and information-theoretic consequences of the representability assumption introduced in Section 4 together with the tensor-network graph-cut bound of Lemma 1. All statements below are conditional on the existence of tensor-network representatives within $\mathcal{M}_{\mathrm{eff}}(D)$.

6.1 Representable Correlation Functionals

Let

$$\rho_k := \left(\Phi_{\mathcal{M}}^{(k)*} \circ \cdots \circ \Phi_{\mathcal{M}}^{(1)*}\right)(\rho_0)$$

denote the sequential Schrödinger evolution induced by manifold-induced channels. Define the intrinsic state on the compressed operator subsystem

$$\omega_k := \rho_k|_{\mathcal{A}_{\mathcal{M}}} \in \mathfrak{S}(\mathcal{A}_{\mathcal{M}}).$$

Under the representability assumption of Section 4, there exists a tensor-network representative

$$\widetilde{\rho}_k \in \mathcal{M}_{\mathrm{eff}}(D)$$

such that

$$\sup_{\substack{X\in\mathcal{A}_{\mathcal{M}}\\ \|X\|\le 1}} |\omega_k(X) - \mathrm{Tr}(X\widetilde{\rho}_k)| \le \eta_D.$$

Whenever a tensor-network representative satisfies the graph-cut Schmidt-rank property of Lemma 1, the corresponding correlation quantities obey the geometric bounds derived below.

Throughout this section, whenever a representative state $\widetilde{\rho}_k \in \mathcal{M}_{\mathrm{eff}}(D)$ is used, we assume that it is realized by a standard tensor-network architecture with bond dimension bounded by D and satisfying the graph-cut Schmidt-rank property of Lemma 1.

6.2 Graph-Cut Capacity Functional

We define the geometric correlation capacity functional

$$\mathcal{C}_G(A,B) := 2\log D \cdot |\partial_G(A,B)|,$$

where $|\partial_G(A,B)|$ denotes the minimal graph cut separating A and B.

This functional depends only on the tensor-network graph structure G and the bond dimension D, and is meaningful only for tensor-network representable states in $\mathcal{M}_{\text{eff}}(D)$. For any pure tensor-network representative $\widetilde{\rho}_k$ satisfying Lemma 1, we have

$$I(A:B)_{\widetilde{\rho}_k} \leq \mathcal{C}_G(A,B).$$

6.3 Scope of the Bounds

The representability hypothesis controls ρ_k only through its restriction to $\mathcal{A}_{\mathcal{M}}$, in the seminorm $\sup_{X\in\mathcal{A}_{\mathcal{M}},\,\|X\|\leq 1} |\operatorname{Tr}(X(\rho_k-\widetilde{\rho}_k))| \leq \eta_D$. The mutual information of ρ_k depends on ρ_k outside $\mathcal{A}_{\mathcal{M}}$, so no bound on $I(A:B)_{\rho_k}$ follows from the hypothesis. All bounds in this section therefore concern the representatives $\widetilde{\rho}_k$.

If one additionally assumes trace-norm closeness, $\|\rho_k-\widetilde{\rho}_k\|_1 \leq \tau_k$, then continuity of the entropy and of the conditional entropy [3, 1] gives $|I(A:B)_{\rho_k} - I(A:B)_{\widetilde{\rho}_k}| \leq f(\tau_k)$, with an explicit f depending on τ_k and the subsystem dimension and vanishing as $\tau_k \to 0$, and hence $I(A:B)_{\rho_k} \leq \mathcal{C}_G(A,B) + f(\tau_k)$. This extra assumption is not implied by representability and is not used elsewhere.

6.4 Sequential Evolution

For each k, the representative $\widetilde{\rho}_k$ satisfies the graph-cut bound of Lemma 1 on its own: $I(A:B)_{\widetilde{\rho}_k} \leq \mathcal{C}_G(A,B)$ for every bipartition $A|B$. This bound does not grow with k, and no cumulative estimate over k is claimed.

Example: Lindbladian realization. The abstract channels $\Phi_{\mathcal{M}}^{(k)} = \mathbb{E}_{\mathcal{M}} \circ \Lambda_k$ may be realized in a variety of physical settings. One important example is obtained by choosing $\Lambda_k = e^{\Delta t\,\mathcal{L}_k}$, where $\mathcal{L}_k$ is a local Lindblad generator [19] acting in the Heisenberg picture. In this case

$$\Phi_{\mathcal{M}}^{(k)}(X) = \mathbb{E}_{\mathcal{M}} \circ e^{\Delta t\,\mathcal{L}_k}(X).$$

The corresponding Schrödinger-picture evolution is given by

$$\Phi_{\mathcal{M}}^{(k)*} = e^{\Delta t\,\mathcal{L}_k^*} \circ \mathbb{E}_{\mathcal{M}}^*.$$

This example illustrates how the abstract manifold-induced channel framework can arise from physically motivated open-system dynamics. The results of the present section, however, do not require a Lindbladian realization and remain valid for arbitrary manifold-induced channels satisfying the assumptions stated above.

6.5 Interpretation

The above results provide a geometric interpretation of correlation propagation under tensor-network representability:

- The graph cut $|\partial_G(A,B)|$ acts as a structural bound on correlation transport.
- The bond dimension D sets the local information throughput across edges.

In this sense, tensor-network geometry imposes an effective constraint on the correlation structure accessible through the compressed operator subsystem $\mathcal{A}_{\mathcal{M}}$.

These statements are conditional on the representability hypothesis: the geometry enters through that hypothesis and is not derived from the operator-algebraic layer.

7 Illustration: Capacity Scaling Across Architectures

7.1 Geometric Scaling of Correlation Capacity

We give a simple comparison illustrating how the geometric capacity functional introduced in Section 6 depends on the underlying tensor-network architecture.

Throughout this example, all statements are understood under the representability assumption of Section 4 together with the tensor-network realization condition stated in Section 6.

Recall that for a tensor-network representative satisfying the graph-cut Schmidt-rank property, the geometric correlation capacity is

$$\mathcal{C}_G(A, B) = 2 \log D \cdot |\partial_G(A, B)|,$$

where D is the bond dimension and $|\partial_G(A, B)|$ denotes the minimal graph cut separating the subsystems A and B.

Under the graph-cut mutual-information bound of Lemma 1, the scaling of the graph cut determines the scaling of the corresponding geometric correlation-capacity functional.

Matrix Product States (MPS). For a one-dimensional open-boundary matrix product state, a left/right bipartition intersects one virtual bond and a contiguous block intersects at most two. Thus

$$|\partial_G(A, B)| = \mathcal{O}(1),$$

and consequently

$$\mathcal{C}_G(A, B) = \mathcal{O}(\log D).$$

The resulting correlation capacity remains independent of system size, reflecting the one-dimensional structure of the MPS geometry.

Projected Entangled Pair States (PEPS). For a two-dimensional PEPS lattice with linear size L, minimal graph cuts scale proportionally to the boundary length of the bipartition:

$$|\partial_G(A, B)| = \mathcal{O}(L).$$

Hence

$$\mathcal{C}_G(A, B) = \mathcal{O}(L \log D).$$

The geometric capacity therefore increases with boundary size, consistent with the familiar area-law structure of two-dimensional tensor networks.

Multiscale Entanglement Renormalization Ansatz (MERA). For MERA architectures, the causal-cone structure implies that effective graph cuts grow only logarithmically with system size:

$$|\partial_G(A, B)| = \mathcal{O}(\log L).$$

Accordingly,

$$\mathcal{C}_G(A, B) = \mathcal{O}(\log L \, \log D).$$

This slower growth reflects the hierarchical multiscale organization of the MERA geometry.

Comparison of geometric scaling. The preceding examples illustrate how different tensor-network architectures induce different geometric correlation capacities.

Tensor Network	Scaling of $\mathcal{C}_G(A, B)$
MPS	$\mathcal{O}(\log D)$
PEPS	$\mathcal{O}(L \log D)$
MERA	$\mathcal{O}(\log L \ \log D)$

The table should be interpreted as a comparison of geometric correlation-capacity scaling under tensor-network representability rather than as a classification theorem.

Interpretation. This example illustrates the central geometric message of the representability framework.

- Bond dimension D controls the local correlation capacity across individual tensor-network bonds.
- Graph-cut size $|\partial_G(A, B)|$ controls the geometric resources available for representing correlations between subsystems.
- Different tensor-network architectures therefore lead to different scaling laws for representable correlations.

In this sense, tensor-network geometry provides a natural geometric language for understanding the correlation structure associated with representable states in the compressed operator framework.

8 Discussion

The present work introduces an operator-algebraic framework for manifold-induced quantum dynamics together with a representability principle connecting compressed operator evolution to tensor-network geometry.

The central structural feature of the framework is the separation between two distinct layers of description.

The first layer is operator-theoretic and is developed in Sections 2 and 3. There, manifold-induced channels are formulated through compressed observable dynamics on operator subsystems associated with a chosen variational manifold. Proposition 1 bounds the difference between step-wise compression and compression applied once at the end.

The second layer is geometric and is introduced through the representability assumption of Section 4. Rather than deriving tensor-network structure directly from the operator-algebraic framework, we assume that intrinsic states arising from the compressed dynamics admit representatives within a class of tensor-network states of bounded bond dimension. Sections 5–7 then explore the consequences of this assumption, showing how graph-cut geometry induces bounds on representable correlations and motivates natural geometric capacity functionals.

A key conceptual contribution of the present work is therefore the identification of a bridge between compressed operator dynamics and tensor-network geometry. In conventional tensor-network approaches, network structure is typically introduced as a variational ansatz for representing many-body quantum states. By contrast, the present framework begins from an operator-system description of dynamics and subsequently investigates the geometric consequences that emerge when representability by tensor-network states is available. This perspective suggests that tensor-network geometry may be understood not only as a representational tool but also as a structural framework for organizing compressed quantum dynamics.

An important feature of the resulting formulation is the logical independence of the two layers. The operator-algebraic results established in Sections 2 and 3 do not depend on tensor-network representability. Conversely, the geometric correlation bounds of Sections 5 and 6 rely only on the representability assumption together with standard tensor-network graph-cut properties. This separation isolates the mathematically rigorous core of the framework from the additional structural assumptions required for geometric interpretation and clarifies precisely where tensor-network information enters the theory.

No universal claim is made that tensor networks provide an exact representation of generic highly entangled states, nor that the derived bounds apply outside the representable sector. The framework is instead a conditional operator-system theory: its dynamical statements are unconditional, while its geometric content is explicitly restricted to states admitting bounded-bond tensor-network representations.

Under the representability assumption, the framework provides a natural geometric interpretation of correlation propagation. Finite bond dimension limits the correlation capacity that can be represented across graph cuts, while the graph structure itself determines how such capacities scale with system architecture. The architecture comparison illustrates that different tensor-network geometries naturally give rise to different correlation-scaling behaviors, reflecting the underlying structure of the corresponding graph cuts. In this sense, tensor-network geometry acts as an organizing principle for representable correlations within the compressed operator framework.

The framework also suggests connections with several active areas of quantum information theory. The manifold-induced channels considered here provide a natural setting for studying adaptive quantum dynamics under restricted representability. The resulting compressed evolutions are closely related to questions of non-Markovian information flow, variational simulation of many-body systems, and operator-theoretic approaches to open-system dynamics. The geometric interpretation developed under representability further connects these questions with tensor-network methods and information-theoretic notions of correlation structure.

At the same time, the present work leaves several important mathematical questions open. Most significantly, the representability assumption is imposed rather than derived. The existence of tensor-network representatives reproducing the intrinsic state structure of the compressed operator subsystem remains a separate problem whose resolution would substantially strengthen the framework. Likewise, questions of uniqueness, stability, and optimality of such representatives remain largely unexplored.

Limitations. Tensor-network geometry (G, D) enters this work as an input, through the admissibility condition on $\mathbb{E}_{\mathcal{M}}$ and the representability hypothesis; the graph-cut bound of Lemma 1 is a standard consequence of those inputs and is not derived from the operator-algebraic layer. No nontrivial admissible compression map is constructed. The assumptions can be satisfied vacuously: for any fixed state ω, $E(X) = \omega(X)\, I$ is unital, completely positive and exactly idempotent, with $\mathcal{A}_{\mathcal{M}} = \mathbb{C}I$, and it satisfies the representability hypothesis with $D = 1$ and $\eta_D = 0$. Whether a linear map E can encode bond-dimension structure nontrivially is open, since the set of operators of bond dimension at most D is not a linear subspace. Finally, Proposition 1 is informative only when the propagated observables stay close to $\mathcal{A}_{\mathcal{M}}$.

A particularly natural direction for future work is the development of a projection theorem establishing the existence of completely positive projections onto tensor-network representable operator subsystems together with explicit approximation guarantees. Such a result would provide a direct mathematical foundation for the representability assumption and would strengthen the geometric interpretation developed in Sections 5 and 6. More broadly, it would establish a fully intrinsic connection between operator-system dynamics and tensor-network geometry, removing the need to postulate representability as an external assumption.

Additional directions for future investigation include the extension of the framework to

fermionic tensor-network manifolds, continuous-variable systems, adaptive manifold-selection procedures, and tensor-network descriptions of open-system control protocols. It would also be interesting to explore whether analogous operator-system constructions exist for more general variational manifolds beyond standard tensor-network architectures.

In summary, the framework developed here provides a mathematically controlled description of manifold-induced dynamics together with a geometric interpretation arising under tensor-network representability. By separating the operator-algebraic and geometric layers of the theory, the construction clarifies both the rigorous foundations of the compressed dynamics and the additional assumptions required for tensor-network interpretations. This separation creates a flexible platform for future investigations at the interface of operator algebras, tensor-network methods, and quantum information theory.

9 Future Directions: Towards an Intrinsic Representability Theory

The framework developed in the present work separates the operator-algebraic construction of manifold-induced dynamics from the geometric consequences that arise under the representability assumption of Section 4. This separation isolates the mathematically rigorous core of the theory while simultaneously identifying a deeper representation problem underlying the connection between compressed operator dynamics and tensor-network geometry.

A natural next objective is therefore the development of an intrinsic representability theory in which the representability assumption emerges as a consequence of the operator-system structure itself. Such a theory would establish a direct mathematical link between manifold-induced channels and tensor-network geometry without requiring representability to be imposed externally.

We briefly outline several interconnected problems that appear central to this program.

9.1 Projection Problem

The representability framework of Section 4 suggests the existence of compression structures that preserve the essential information contained within a restricted tensor-network description.

A fundamental question is whether one can construct completely positive maps

$$\Pi_D : \mathcal{B}(\mathcal{H}) \longrightarrow \mathcal{A}_D,$$

where $(\mathcal{A}_D)$ denotes an operator subsystem associated with tensor-network representations of bond dimension (D), such that:

- Π_D is completely positive and unital;
- Π_D is approximately idempotent;
- the approximation error admits a controlled scaling

$$\varepsilon(D) \to 0 \qquad (D \to \infty).$$

More precisely, one may seek estimates of the form

$$||\Pi_D^2 - \Pi_D||_\diamond \leq \varepsilon(D),$$

together with quantitative control of the induced compression error.

The existence of such projections would provide a direct mathematical foundation for the representability assumption introduced in Section 4 and would establish a canonical operator-system structure associated with finite-bond tensor-network descriptions.

9.2 Operator-System Closure Problem

Assuming suitable projection maps exist, a second question concerns the intrinsic structure of the resulting compressed observables.

Specifically, one may ask under what conditions tensor-network representable observables generate operator subsystems

$$\mathcal{A}_D \subseteq \mathcal{B}(\mathcal{H})$$

that are stable under the compressed dynamics.

Possible directions include characterizing when such subsystems admit:

- operator-system structures in the sense of unital self-adjoint subspaces;
- approximate closure under manifold-induced channels;
- approximate conditional expectation properties;
- stability under sequential adaptive compression.

Understanding these properties would place tensor-network representability within a more systematic operator-algebraic framework and could clarify the extent to which compressed dynamics admits an intrinsic description independent of any particular tensor-network parametrization.

9.3 Geometric Reconstruction Problem

A complementary question concerns reconstruction of tensor-network states from intrinsic operator-system data.

Given a state

$$\omega \in \mathfrak{S}(\mathcal{A}_D),$$

one seeks conditions guaranteeing the existence of a tensor-network state

$$\widetilde{\rho} \in \mathcal{M}_{\mathrm{eff}}(D)$$

satisfying

$$\omega = \widetilde{\rho}|_{\mathcal{A}_D}.$$

More generally, one may ask whether such representatives can be chosen in a stable or approximately unique manner and how reconstruction errors scale with bond dimension.

A solution to this problem would provide a direct realization of intrinsic operator-system states as tensor-network states and would place the representability assumption of Section 4 on a constructive footing.

9.4 Towards an Intrinsic Geometric Theory

The three problems described above are closely related and should not be viewed as independent directions.

The projection problem addresses the existence of compression structures compatible with tensor-network geometry. The operator-system closure problem seeks to characterize the resulting observable algebras. The geometric reconstruction problem concerns recovery of tensor-network states from the intrinsic state space of those algebras.

Taken together, these questions suggest a broader program in which tensor-network geometry emerges directly from operator-system dynamics rather than being imposed as an external representational layer.

A successful resolution of these problems would replace the representability assumption of Section 4 by an intrinsic mathematical construction. The geometric correlation bounds of Sections 5 and 6 would then become derived consequences of the compressed dynamics itself, yielding a fully internal connection between manifold-induced channels, operator-system geometry, and tensor-network representations.

Possible mathematical approaches. The problems outlined above naturally connect the present framework to several established areas of operator theory and quantum information. Potential technical tools include operator-system theory, Choi-matrix characterizations of completely positive maps, approximate conditional expectations in finite-dimensional operator algebras, and geometric reconstruction methods for tensor-network state manifolds.

Developing these connections may lead to an intrinsic operator-algebraic characterization of tensor-network representability and provide a rigorous foundation for the projection and reconstruction structures assumed in the present work.

Beyond these foundational questions, it would be interesting to extend the framework to fermionic tensor-network manifolds, continuous-variable systems, adaptive manifold-selection schemes, and operator-algebraic formulations of variational quantum dynamics. Such developments may help clarify the broader role of compressed operator structures in many-body quantum information theory and open-system dynamics.

10 Conclusion

We have introduced a framework in which a unital completely positive compression map $\mathbb{E}_{\mathcal{M}}$, whose image is an operator system $\mathcal{A}_{\mathcal{M}}$ labeled by a tensor-network geometry (G, D), induces Heisenberg-picture channels $\Phi_{\mathcal{M}} = \mathbb{E}_{\mathcal{M}} \circ \Lambda$. The resulting channel class is exactly closed under composition (Observation 1), and Proposition 1 bounds the difference between step-wise compression and compression applied once at the end, pointwise on propagated observables.

Tensor-network geometry enters through the admissibility condition on $\mathbb{E}_{\mathcal{M}}$ and through a representability hypothesis on $\mathcal{A}_{\mathcal{M}}$. Under that hypothesis the standard Schmidt-rank argument bounds the mutual information of the representatives by the graph-cut capacity $2 \log D \, |\partial_G(A, B)|$ (Lemma 1), and Sec. 7 compares this capacity across MPS, PEPS and MERA geometries. These bounds are conditional: the geometry is not derived from the operator-algebraic layer, and no nontrivial admissible compression map is constructed here.

A key structural feature of the framework is the separation between an operator-algebraic layer describing compressed dynamics and a conditional geometric layer arising under tensor-network representability.

Finally, the analysis identifies a broader research program centered on intrinsic representability. Establishing completely positive projection structures, characterizing tensor-network representable operator systems, and developing geometric reconstruction maps would provide a fully intrinsic operator-algebraic foundation for tensor-network channel dynamics. We hope that the framework developed here serves as a first step toward such a theory and stimulates further interaction between tensor-network methods, operator algebras, and quantum information theory.

A Proof of Proposition 1

Proof. Write $E := \mathbb{E}_{\mathcal{M}}$. For $j = 1, \ldots, n$ define

$$A_j := E\Lambda_n \, E\Lambda_{n-1} \cdots E\Lambda_j \, \Lambda_{j-1} \cdots \Lambda_1.$$

Then $A_1 = \Phi_{\mathcal{M}}^{(n)} \circ \cdots \circ \Phi_{\mathcal{M}}^{(1)}$ and $A_n = E\Lambda_n\Lambda_{n-1}\cdots\Lambda_1$, so $\Delta_n = \|A_1 - A_n\|_\diamond$. For $1 \le j \le n-1$ put

$$P_j := E\Lambda_n \cdots E\Lambda_{j+1}, \qquad Q_j := \Lambda_{j-1}\cdots\Lambda_1$$

($Q_1 = \mathrm{id}$); both are unital completely positive. Then

$$A_j - A_{j+1} = P_j\,(E\Lambda_j - \Lambda_j)\,Q_j, \qquad A_1 - A_n = \sum_{j=1}^{n-1}(A_j - A_{j+1}).$$

Unital completely positive maps are contractions in operator norm, so for every X,

$$\|(A_j - A_{j+1})(X)\| \le \|(\mathrm{id} - E)(\Lambda_j Q_j X)\| = \|(\mathrm{id} - E)(\Lambda_j \cdots \Lambda_1 X)\|,$$

and summing over j gives the pointwise bound. In diamond norm, submultiplicativity and $\|P_j\|_\diamond = \|Q_j\|_\diamond = 1$ give

$$\|A_j - A_{j+1}\|_\diamond \le \|\Lambda_j - E\Lambda_j\|_\diamond = d_{\mathcal{M}}(\Lambda_j),$$

whence $\Delta_n \le \sum_{j=1}^{n-1} d_{\mathcal{M}}(\Lambda_j)$. Idempotence of E is not used. □

B Proof of Lemma 1: Graph-Cut Mutual Information Bound

Proof. **Step 1: Schmidt-rank bound from the graph cut.** We consider a pure state $|\psi\rangle \in \mathcal{M}$; all rank and entropy arguments below assume purity. Consider a minimal graph cut separating A from B. Let the cut intersect the set of bonds $\{e_1, e_2, \ldots, e_{|\partial_G(A,B)|}\}$, each with bond dimension at most D. Across the bipartition $A|B$, every cut bond contributes at most a factor D to the Schmidt rank. Since the minimal cut intersects $|\partial_G(A,B)|$ bonds, the Schmidt rank satisfies

$$\chi(A|B) \le D^{|\partial_G(A,B)|}.$$

Equivalently, the reduced density matrix ρ_A satisfies

$$\mathrm{rank}(\rho_A) \le D^{|\partial_G(A,B)|}.$$

Step 2: Entropy bound. Using $S(\rho) \le \log \mathrm{rank}(\rho)$, we obtain

$$S(\rho_A) \le |\partial_G(A,B)| \log D,$$

and similarly

$$S(\rho_B) \le |\partial_G(A,B)| \log D.$$

For a pure state $|\psi\rangle$, we have $S(\rho_{AB}) = 0$, hence

$$I(A:B)_\psi = S(\rho_A) + S(\rho_B) \le 2 \log D \cdot |\partial_G(A,B)|.$$

□

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